Fair Division in India's Public Distribution System:

A Multi-Rule Analysis of Two-Good Fair Allocation of Rice and Wheat under Heterogeneous Preferences

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Abstract

India's Public Distribution System (PDS) allocates subsidised rice and wheat to over 800 million beneficiaries, with state-level quantities fixed by the National Food Security Act (NFSA) 2013. This report models the PDS grain allocation problem as a two-good fair division instance across $N = 36$ states and Union Territories. Using MPI-based poverty headcounts and HCES 2022–23 consumption preferences, we evaluate four allocation rules—Proportional (PRO), Weighted Nash Social Welfare (wNSW), Atkinson $\varepsilon = 2$, and Leximin—against the NFSA baseline. We show that the Leximin rule yields an analytically characterised equal-utility solution (reduced to a one-dimensional root-finding problem), derive the worst-off state under wNSW analytically, and establish that PRO is envy-free under the scaled definition while wNSW, Atkinson, and Leximin are not. Six axiomatic properties are formally verified using Z3 SMT: PRO monotonicity, scale invariance of PRO and wNSW, PRO envy-freeness, the wNSW floor convexity, and the a^* formula; envy-freeness of wNSW and Leximin is refuted with explicit polynomial witnesses. On the comparable subset of 16 states with two-good NFSA baselines, four states (Bihar, Uttar Pradesh, Jharkhand, D&NH) robustly improve under all rules.

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1. Introduction

The National Food Security Act (NFSA) 2013 entitles priority households to 5 kg of foodgrains per person per month at heavily subsidised prices. Allocations to states are fixed by the central government using 2011 Census population estimates, creating a mismatch with current poverty distribution. As India's poverty landscape has shifted, documented by the 2021 MPI and 2024 Aadhaar-based population estimates and some high-poverty states receive less grain per poor person than historically better-off states. In the data used here, the correlation between a state's MPI headcount ratio and its NFSA allocation per poor person is 0.50: Bihar (H = 0.338) receives 127 kg per poor person per year, while Kerala (H = 0.005) receives 5,191 kg — a 40-fold gap driven by the historical beneficiary counts rather than current poverty.

This project asks: given a fixed annual supply of rice (M_R = 35,900.92 kT) and wheat (M_W = 16,210.90 kT), how should these be allocated across states to satisfy normative criteria such as proportionality, aggregate welfare maximisation, inequality aversion, and equitable treatment of the worst-off? We compare four allocation rules against the NFSA baseline, characterise their fairness properties formally, and use Z3 SMT verification to check axiomatic claims.

The primary contribution is the application of classical fair division theory typically studied in abstract settings to a concrete, data-driven policy problem where heterogeneous

preferences (rice vs. wheat preference) interact with heterogeneous poverty rates to determine which rule best serves India's food security goals.

2. Background

2.1 PDS Structure and NFSA

The PDS operates through a network of Fair Price Shops. Under NFSA, state allocations are the product of the number of beneficiaries (Antyodaya Anna Yojana + Priority Household categories) and per-capita entitlements. Coarse grains (jowar, bajra) are excluded from this analysis; only rice and wheat are modelled. Two states, Chandigarh and Puducherry, have migrated to Direct Benefit Transfer (DBT) cash payments and receive zero in-kind NFSA allocation despite having nonzero poor populations.

2.2 Poverty Measurement

We use the MPI headcount ratio H_i from the 2021 MPI India report (OPHI, 2021) as our measure of poverty intensity. Poor population is $n_i = H_i \times \text{Pop}_i$, where Pop_i is the 2024 Aadhaar Saturation Report population. Total poor population across 36 states is $\sum n_i = 211.26$ million. This is substantially smaller than NFSA's 813 million beneficiaries, reflecting the MPI's multidimensional poverty threshold versus NFSA's broader inclusion criteria.

2.3 Consumption Preferences

Rice-wheat preference is estimated from HCES 2022–23, Table 3.8R (rural consumption). We define $a_i \in (0, 1)$ as the rice share of total cereal consumption:

$$a_i = \frac{\text{Rice consumption}_i}{\text{Rice consumption}_i + \text{Wheat consumption}_i}$$

Direct estimates were available for 18 states; for the remaining 18 (mostly Northeast and UTs), we used regional proxies. Values range from $a_i = 0.043$ (Rajasthan) to $a_i = 0.954$ (Manipur), reflecting well-known regional dietary patterns.

3. Model

3.1 Problem Formulation

We have $N = 36$ agents (states), two goods (rice, wheat), with fixed supplies M_R and M_W. An allocation is $(\mathbf{x}^R, \mathbf{x}^W) \in \mathbb{R}^N_{>0} \times \mathbb{R}^N_{>0}$ satisfying:

$$\sum_{i=1}^{N} x_i^R = M_R, \qquad \sum_{i=1}^{N} x_i^W = M_W$$

3.2 Welfare Metric

Each state i has n_i poor persons with Cobb-Douglas preferences over per-capita grain bundles. Let $r_i = x_i^R/n_i$ and $w_i = x_i^W/n_i$ denote per-capita allocations, and let $\bar{r}$ denote the unit in which r_i and w_i are expressed (so $r_i/\bar{r}$ and $w_i/\bar{r}$ are dimensionless real numbers). The per-capita welfare of state i is:

$$\tilde{u}_i = a_i \log\left(\frac{r_i}{\bar{r}}\right) + (1 - a_i) \log\left(\frac{w_i}{\bar{r}}\right) \tag{1}$$

The reference unit $\bar{r}$ enters as $-(a_i + (1-a_i)) \log \bar{r} = -\log \bar{r}$, an additive constant identical for all states. Consequently: **(i)** utility *differences* $\tilde{u}_i - \tilde{u}_j$ are fully unit-invariant; **(ii)** the Atkinson index is computed on $u_i = \exp(\tilde{u}_i)$ and equals a ratio of means, hence unit-invariant; **(iii)** aggregate welfare $\sum n_i \tilde{u}_i$ shifts by $-\log \bar{r}$ under unit change and is therefore meaningful only as *cross-rule comparisons at fixed units*, not as absolute quantities. All comparisons in this report are of the form "rule A vs rule B at fixed supply," for which (i) holds. The absolute values $\tilde{u}_i \approx -9$ reflect the kT/person scale and carry no independent interpretation.

This is the log of a Cobb-Douglas utility function $u_i = (r_i)^{a_i}(w_i)^{1-a_i}$ where $r_i = x_i^R/n_i$ and $w_i = x_i^W/n_i$ are per-capita allocations. Cobb-Douglas has an elasticity of substitution of exactly one. It penalises bundles that deviate from the preference ratio $a_i/(1 - a_i)$, which is appropriate for PDS modelling since rice and wheat satisfy nutritionally distinct dietary needs across regions.

The aggregate social welfare is $W = \sum_{i=1}^{N} n_i \tilde{u}_i$ (population-weighted sum).

3.3 Allocation Rules

PRO (Proportional). Each state receives a fraction of total supply proportional to its poor population:

$$x_i^{R,\text{PRO}} = \frac{n_i}{\sum_j n_j} M_R, \qquad x_i^{W,\text{PRO}} = \frac{n_i}{\sum_j n_j} M_W \tag{2}$$

This yields identical per-capita allocations for all states: $r_i \approx 170$ kg/person/yr and $w_i \approx 77$ kg/person/yr.

wNSW (Weighted Nash Social Welfare). Maximises the population-weighted sum of log utilities:

$$\max_{\mathbf{x}} \sum_{i=1}^{N} n_i \tilde{u}_i = \max_{\mathbf{x}} \log \prod_i u_i^{n_i}$$

The problem separates over goods due to the additive structure of $\tilde{u}_i$. The KKT conditions yield:

$$x_i^{R,\text{wNSW}} = \frac{n_i a_i}{\sum_j n_j a_j} M_R, \qquad x_i^{W,\text{wNSW}} = \frac{n_i(1 - a_i)}{\sum_j n_j (1 - a_j)} M_W \tag{3}$$

This is distinct from the unweighted Nash product $\prod u_i$. The population weights reflect that a state represents n_i individuals. Each state receives grain proportional to its preference-weighted poor population.

Atkinson $\varepsilon = 2$. The standard Atkinson social welfare function with parameter ε is $W = \frac{1}{1-\varepsilon}\sum_i u_i^{1-\varepsilon}$ (up to a sign). For $\varepsilon = 2$ this gives $W \propto -\sum_i 1/u_i$. We adopt a *population-weighted* version so that each poor individual is weighted equally rather than each state:

$$\max_{\mathbf{x}} \left(-\sum_{i=1}^{N} \frac{n_i}{u_i} \right) \equiv \min_{\mathbf{x}} \sum_{i=1}^{N} \frac{n_i}{u_i}$$

Unlike wNSW, which maximises aggregate welfare, and Leximin, which equalises utilities exactly, the Atkinson objective imposes a strong but smooth penalty on low-utility states (via the $1/u_i$ term), producing an intermediate allocation with a lower floor than Leximin but less inequality than wNSW.

Log-space reformulation. Since $u_i = r_i^{a_i} w_i^{1-a_i}$, we have $1/u_i = r_i^{-a_i} w_i^{-(1-a_i)}$. Setting $p_i = \log r_i$ and $q_i = \log w_i$:

$$\frac{1}{u_i} = e^{-(a_i p_i + (1-a_i) q_i)}$$

The problem therefore becomes:

$$\min_{p,\,q} \sum_{i=1}^{N} n_i \exp\bigl(-(a_i p_i + (1-a_i) q_i)\bigr) \tag{4}$$

subject to $\sum_i n_i e^{p_i} = M_R$ and $\sum_i n_i e^{q_i} = M_W$.

KKT structure. The first-order conditions with multipliers λ_R, λ_W are:

$$\begin{aligned} -n_i a_i \exp\bigl(-(a_i p_i + (1-a_i) q_i)\bigr) + \lambda_R n_i e^{p_i} &= 0 \\ -n_i (1-a_i) \exp\bigl(-(a_i p_i + (1-a_i) q_i)\bigr) + \lambda_W n_i e^{q_i} &= 0 \end{aligned}$$

Dividing the two equations: $e^{p_i - q_i} = \frac{a_i \lambda_W}{(1-a_i)\lambda_R}$, implying $r_i/w_i \propto a_i/(1-a_i)$ at optimum — the grain ratio is aligned with each state's preference ratio, qualitatively similar to wNSW but with inequality-averse weights.

Convexity and numerical solution. The objective in (4) is a sum of convex functions (exponentials of linear terms), hence convex. The equality constraints $\sum_i n_i e^{p_i} = M_R$ and $\sum_i n_i e^{q_i} = M_W$ are nonlinear, so this is not a standard convex program and global optimality is not guaranteed by KKT alone. We solve via SLSQP with the wNSW solution as initialisation. Multiple random restarts yielded identical solutions; primal feasibility residuals were below 10^{-8} kT, confirming convergence to a consistent optimum.

Leximin. Maximises the minimum $\tilde{u}_i$, then the second minimum, and so on lexicographically (Rawlsian maximin principle).

The equal-utility condition $\tilde{u}_i = c$ for all i, together with the KKT conditions of the supply constraints (shadow prices λ_R, λ_W), yields the allocation formulas: setting $k = \lambda_R/\lambda_W$,

$$r_i = \left(\frac{a_i}{k(1-a_i)}\right)^{1-a_i} e^c, \qquad w_i = \left(\frac{k(1-a_i)}{a_i}\right)^{a_i} e^c \tag{5}$$

Substituting into the supply constraints and dividing one by the other gives the one-dimensional equation for k:

$$f(k) \;=\; \frac{\sum_i n_i \left(\frac{a_i}{k(1-a_i)}\right)^{1-a_i}}{\sum_i n_i \left(\frac{k(1-a_i)}{a_i}\right)^{a_i}} - \frac{M_R}{M_W} = 0 \tag{6}$$

Proposition 1 (Monotonicity of f)**.** *The function $f(k)$ is strictly decreasing in $k > 0$.*

Proof. Write $f(k) = G(k)/H(k) - M_R/M_W$ where $G(k) = \sum_i n_i(a_i/(k(1-a_i)))^{1-a_i}$ and $H(k) = \sum_i n_i(k(1-a_i)/a_i)^{a_i}$. Differentiating and simplifying: $G'(k) = -(1/k)\sum_i n_i(1-a_i)^2(a_i/(k(1-a_i)))^{1-a_i} < 0$ and $H'(k) = (1/k)\sum_i n_i a_i(1-a_i)(k(1-a_i)/a_i)^{a_i} > 0$. Since $G' < 0$, $H' > 0$, $G > 0$, $H > 0$: $(G/H)' = (G'H - GH')/H^2 < 0$. Hence f is strictly decreasing with a unique root, found via Brent's method. Numerically: $k^* = 0.3605$, $\tilde{u}^* = -8.9788$. □

Theorem 1 (Leximin = Equal Utility)**.** *Let $a_i \in (0,1)$ and $n_i > 0$ for all i. Then the unique leximin allocation satisfies $\tilde{u}_i = \tilde{u}^*$ for all i.*

Proof. **Step 1 (Existence).** The feasible set

$$\mathcal{X} = \left\{\mathbf{x} \in \mathbb{R}^{2N}_{>0} : \textstyle\sum_{i=1}^N x_i^R = M_R,\ \sum_{i=1}^N x_i^W = M_W\right\}$$

is the relative interior of a compact simplex face. Taking its closure $\overline{\mathcal{X}}$ (which includes boundary points $x_i^R = 0$ or $x_i^W = 0$) yields a compact set. The function $\min_i \tilde{u}_i(\mathbf{x})$ is continuous on $\mathcal{X}$ and, since $a_i \in (0,1)$, equals $-\infty$ on $\partial\mathcal{X}$. Therefore the supremum is attained in the interior: a maximin solution $\mathbf{x}^* \in \mathcal{X}$ exists by the Extreme Value Theorem.

Step 2 (Equal utilities). Let $t^* = \min_i \tilde{u}_i(\mathbf{x}^*)$. Suppose for contradiction that $\tilde{u}_k(\mathbf{x}^*) > t^*$ for some k, and let j satisfy $\tilde{u}_j(\mathbf{x}^*) = t^*$. Since $\tilde{u}_k$ is strictly increasing and continuous in (x_k^R, x_k^W), there exists $\varepsilon > 0$ small enough that transferring $(\varepsilon, \varepsilon)$ from agent k to agent j satisfies: (i) $\tilde{u}_k$ remains strictly above t^* (by continuity); (ii) $\tilde{u}_j$ strictly increases (by strict monotonicity). The new minimum is therefore strictly greater than t^*, contradicting the optimality of $\mathbf{x}^*$. Hence $\tilde{u}_i(\mathbf{x}^*) = t^*$ for all i.

Step 3 (Uniqueness). The equal-utility condition $\tilde{u}_i = c$ for all i parametrises the allocation by (k, c) via Equations (5). The supply constraints pin k uniquely via $f(k^*) = 0$ (Proposition 1), and then $e^c = M_R/\sum_i n_i r_i(k^*)$ is uniquely determined. Hence the equal-utility allocation is unique.

Step 4 (Leximin optimality). Since the maximin solution is unique, there is no scope for further lexicographic improvement. The unique maximin allocation therefore coincides with the leximin allocation. □

4. Data

All data is publicly available. Table 1 summarises sources and construction decisions.

Table 1: Data sources and construction

Variable	Source and Notes
n_i (poor population)	MPI India 2021 (OPHI, 2021) headcount × 2024 Aadhaar population. Mixed reference years acknowledged.
M_R, M_W (supply)	FCI April 2025 allocation bulletin, AAY + PHH categories, ×12 for annual. Coarse grains excluded.
a_i (rice preference)	HCES 2022–23 Table 3.8R (rural). Direct for 18 states; regional proxies for 17 others. Five NE states (Arunachal Pradesh, Manipur, Meghalaya, Mizoram, Nagaland) use Assam's value $a = 0.9537$; Sikkim and Tripura use $a = 0.8123$; Lakshadweep uses $a = 0.8758$; remaining UTs and hill states use proxies from neighbouring regions. Note: Arunachal Pradesh has documented wheat flour and noodle consumption that the Assam proxy overstates rice preference for; state-level allocations for proxied states are accordingly sensitive.
DBT states	Chandigarh ($n = 43{,}754$, $a = 0.094$) and Puducherry ($n = 11{,}825$, $a = 0.927$) have zero NFSA allocation. Included in Case A; excluded in Case B. Results nearly identical ($< 0.026\%$ of total poor).

4.1 Acknowledged Limitations

- a_i is estimated from rural HCES only; urban consumption is excluded. This is a notable source of bias for Delhi and Maharashtra.
- NFSA beneficiary count (813M) is far larger than MPI-poor count (211M). The welfare comparison with NFSA is therefore imperfect, particularly for low-H_i states where NFSA allocations are driven by political rather than poverty factors.
- a_i is treated as fixed regardless of the allocation; no demand elasticity is modelled.

5. Results

5.1 Fairness Metrics

Table 2 reports corrected fairness metrics for Case A (all 36 states). Gini coefficient was excluded as it is not unit-invariant for log-scale utilities; shifting $\tilde{u}_i$ by a constant changes the Gini. We use unweighted std($\tilde{u}_i$) and the Atkinson index on $u_i = \exp(\tilde{u}_i)$ instead.

† Denominator is 16 (not 36): 20 states receive only a single grain type under NFSA (rice = 0 or wheat = 0), making log-CD utility undefined for their NFSA baseline. All 36 states are included in the allocation rules.

Table 2: Fairness metrics, Case A (all 36 states/UTs)

Metric	**PRO**	**wNSW**	**Atk.**	**Lex.**
min $\tilde{u}_i$ (worst-off floor)	−9.4407	−9.1522	−9.0660	−8.9788
max $\tilde{u}_i$ (best-off)	−8.7169	−8.2155	−8.5584	−8.9788
range $\tilde{u}_i$	0.7238	0.9367	0.5076	0.0000
std $\tilde{u}_i$ (unweighted)	0.2599	0.3557	0.1954	0.0000
pop-weighted std $\tilde{u}_i$	0.1964	0.2957	0.1602	0.0000
$\sum n_i \tilde{u}_i$ ($\times 10^9$)	−1.917	−1.888	−1.890	−1.897
Atkinson index ($\varepsilon = 2$) on u_i	0.0649	0.1171	0.0373	0.0000
pop-weighted Atkinson	0.0378	0.0840	0.0253	0.0000
floor Δ vs NFSA worst-off	0.4757	0.7641	0.8504	0.9376
states better than NFSA†	5/16	5/16	5/16	5/16
poor pop. better than NFSA (M)	127.65	127.65	127.65	127.65

5.2 Key Findings

Finding 1: Five states improve under every rule (on the comparable subset). Among the 16 states with valid NFSA baselines, Bihar, Uttar Pradesh, Madhya Pradesh, Jharkhand, and D&NH improve relative to NFSA under all four rules. Table 2 reports this as 5/16.

Denominator mismatch and robustness. NFSA allocations were sized for the 2011 Census population; our rules use 2024 Aadhaar-based poor population counts. Using the larger 2024 denominator lowers the NFSA per-capita figure below what beneficiaries actually received. To bound the effect: high-poverty states in this group grew at roughly 10–15% over 2011–2024, implying NFSA per-capita was overstated (in the 2024 framework) by approximately $\log(1.15) \approx 0.14$ utility units. After applying this correction to the NFSA baseline, Bihar (gap 0.65), Jharkhand (0.38), D&NH (0.56), and Uttar Pradesh (0.06) remain improvers under PRO, the most conservative rule. Madhya Pradesh (corrected gap ≈ -0.04 under PRO) is the borderline case: its improvement result is sensitive to the population denominator. The finding is best stated as: four states improve robustly under all rules; Madhya Pradesh improves under the data as presented but is a marginal case subject to the 2011/2024 mismatch.

Finding 2: PRO disadvantages wheat-preferring states. Under PRO, all states receive $r = 170$ kg and $w = 77$ kg per poor person per year, giving a fixed ratio $M_R/M_W = 2.215$. States with low a_i (strong wheat preference, e.g., Rajasthan $a = 0.043$, Haryana $a = 0.080$) are structurally disadvantaged: the Cobb-Douglas utility is maximised when the ratio of allocated goods aligns with the preference ratio $a_i/(1 - a_i)$. A uniform bundle ignores this.

Finding 3: wNSW worst-off states are those near $a^* = 0.313$. Under wNSW, the shadow prices of the two supply constraints are $C_R = M_R/\sum_j n_j a_j$ and $C_W = M_W/\sum_j n_j(1-a_j)$. These are system-level constants determined by aggregate preferences

and total supply, independent of any individual state's a_i. Treating C_R and C_W as fixed, the welfare of state i under wNSW is:

$$\tilde{u}_i^{\text{wNSW}} = a_i \log(a_i C_R) + (1 - a_i) \log\big((1 - a_i) C_W\big)$$

Differentiating with respect to a_i:

$$\frac{d\tilde{u}_i}{da_i} = \log\left(\frac{a_i C_R}{(1 - a_i) C_W}\right)$$

This derivative is zero at $a_i = C_W/(C_R + C_W)$ and the second derivative is $d^2\tilde{u}_i/da_i^2 = 1/a_i + 1/(1 - a_i) > 0$ for all $a_i \in (0, 1)$. Hence $\tilde{u}_i^{\text{wNSW}}$ is strictly *convex* in a_i, with a unique *minimum* at:

$$a^* = \frac{C_W}{C_R + C_W}$$

Using the computed values $C_R = M_R/\sum_j n_j a_j$ and $C_W = M_W/\sum_j n_j(1 - a_j)$, we obtain $a^* = 0.313$. States with $a_i \approx 0.313$—Gujarat ($a = 0.329$), D&NH ($a = 0.329$), Madhya Pradesh ($a = 0.276$)—are worst-off under wNSW. Their marginal rate of substitution equals the system-wide shadow price ratio, so wNSW's preference-tailoring offers them no relative advantage. States far from a^* in either direction receive allocations better matched to their preferences.

Finding 4: Leximin tailors grain strongly to preference. With $k^* = 0.3605$, rice-preferring states (high a_i) receive mostly rice and very little wheat; wheat-preferring states receive the reverse. Selected per-capita allocations under Leximin:

Table 3: Leximin per-capita allocation (selected states)

State	a_i	Rice (kg/p/yr)	Wheat (kg/p/yr)	Total
Rajasthan	0.043	17	138	155
Haryana	0.080	34	141	175
Bihar	0.526	84	59	143
Kerala	0.876	138	14	152
Manipur	0.954	152	3	155

5.3 Welfare–Inequality Trade-off

The four rules trace a trade-off: wNSW achieves the highest aggregate welfare ($\sum n_i \tilde{u}_i = -1.888 \times 10^9$) but the highest inequality (Atkinson index 0.1171). Leximin achieves perfect equality (Atkinson = 0) but lower aggregate welfare (-1.897×10^9). Atkinson $\varepsilon = 2$ sits between them in both dimensions. PRO is weakly dominated-lower welfare than wNSW and higher inequality than Atkinson.

6. Axiomatic Verification via Z3 SMT

All properties below were verified using Z3 version 4.16.0 (z3-solver 4.16.0 Python bindings).

6.1 Approach

Axiomatic properties are verified symbolically using Z30. For properties involving the utility function $u_i = r_i^{a_i} w_i^{1-a_i}$, we exploit the fact that when $a_i = p/q$ is rational, raising both sides to the q-th power converts the expression to a polynomial. For instance, $a_i = 13/16 \approx 0.8125$ (Andaman & Nicobar Islands, error < 0.001) gives $u_i^{16} = r_i^{13} w_i^3$. We basically rationalise exponents and convert them into polynomial inequalities, which Z3 can handle exactly.

6.2 Results

Table 4: Axiomatic verification results (Z3 v4.16.0). P1–P2 are algebraic verifications of closed-form expressions; P3–P4 are deeper SMT checks over nonlinear arithmetic.

Property	**Rule(s)**	**Result**	**Notes**
P1: Monotonicity ($n_i \uparrow \Rightarrow x_i \uparrow$)	PRO	**UNSAT**	Proved [13ms]; algebraic
P2: Scale invariance	PRO, wNSW	**UNSAT**	Proved [0.1ms each]; algebraic identity on closed form
P2: Scale invariance	Atkinson, Leximin	*[Pending]*	Optimizer-dependent
P3: Envy-freeness	PRO	**UNSAT**	Proved [0.1ms]; reflexivity on identical per-capita bundles
P3: Envy-freeness	wNSW	**SAT**	Witness: Odisha envies CG, $a \approx 73/80$ [5ms]
P3: Envy-freeness	Atkinson	**SAT**	Witness: Kerala envies Jharkhand, $a \approx 7/8$ [0.1ms]
P3: Envy-freeness	Leximin	**SAT**	Witness: A&N envies Bihar, $a \approx 13/16$ [0.2ms]
P4: wNSW convexity ($d^2\tilde{u}/da^2 > 0$)	wNSW	**UNSAT**	Proved [7ms]
P4: wNSW minimum $a^* = C_W/(C_R+C_W)$	wNSW	**UNSAT**	Proved [6ms]

6.3 Selected Proof Traces

P1 — PRO Monotonicity (UNSAT, algebraic). Under PRO, $x_i^R = n_i M_R/(n_i + S)$ where $S = \sum_{j \neq i} n_j$. We verify algebraically that for all $n_i, S, \delta > 0$:

$$\frac{n_i + \delta}{n_i + \delta + S} > \frac{n_i}{n_i + S}$$

Z3 asserts $n_i, S, \delta > 0$ and the negation; returns UNSAT in 13ms. This is an algebraic check on the PRO closed form, not a statement about monotonicity of the underlying optimisation.

P2 — Scale Invariance (UNSAT, algebraic identity). PRO and wNSW allocations are linear in M_R and M_W by their closed forms ($x_i^{\text{PRO}} = (n_i/\sum n_j)M_R$, $x_i^{\text{wNSW}} = (n_i a_i/\sum n_j a_j)M_R$). Z3 verifies $(n_i/N)(\lambda M) = \lambda(n_i/N)M$ as a polynomial identity in 0.1ms. These checks confirm the algebraic structure of the closed forms; scale invariance of Atkinson and Leximin (which are defined by optimisation, not closed forms) remains pending.

P3a — PRO Envy-Freeness (UNSAT). Under PRO, every state receives the same per-capita bundle (r, w). The envy condition $u_i(r_j, w_j) > u_i(r_i, w_i)$ becomes $u_i(r, w) > u_i(r, w)$, which is unsatisfiable by reflexivity. Z3 confirms in 0.1ms. This is a consequence of the identical per-capita bundles under PRO, not a deep SMT result.

P3b — wNSW Envy (SAT, Odisha envies Chhattisgarh). Rational approximation: $a_{\text{Odisha}} = 0.9135 \approx 73/80$; error $= 73/80 - 0.9135 = 0.0010$. Raising the utility comparison to the 80th power gives the integer inequality $3095^{73} \times 131^7 > 3090^{73} \times 133^7$ (per-capita values $\times 10$). Z3 returns SAT in 5ms with integer witnesses. Under wNSW, Chhattisgarh ($a = 0.9150$) receives slightly more rice per capita and slightly less wheat; evaluated under Odisha's preferences ($a = 0.9135$), this bundle is strictly preferred.

P3c — Atkinson Envy (SAT, Kerala envies Jharkhand). $a_{\text{Kerala}} = 0.8758 \approx 7/8$; error $= 0.0008 < 0.001$. Polynomial witness (8th power): $2316^7 \times 439 > 2406^7 \times 138$ (per-capita $\times 10$). Z3 returns SAT in 0.1ms. Under Atkinson, Jharkhand receives more wheat than Kerala (43.9 vs. 13.8 kg/p/yr) while having slightly less rice (231.6 vs. 240.6); evaluated under Kerala's rice-heavy preferences ($a = 0.8758$), the extra wheat in Jharkhand's bundle still outweighs the rice deficit.

P3d — Leximin Envy (SAT, A&N envies Bihar). $a_{\text{A\&N}} = 0.8123 \approx 13/16$; error $= 0.0002 < 0.001$. Polynomial witness (16th power): $2147^{13} \times 698^3 > 2010^{13} \times 167^3$ (per-capita $\times 10$). Z3 returns SAT in 0.2ms. Bihar's higher absolute rice allocation (214.7 vs. 201.0 kg/p/yr) is preferred by A&N's rice-heavy preferences, even though both states have equal $\tilde{u}_i = -8.9788$.

P4 — wNSW Floor Convexity (UNSAT). Z3 asserts $a \in (0, 1)$ and the negation $1/a + 1/(1-a) \leq 0$, returning UNSAT in 7ms. After clearing denominators, the negation requires $1/(a(1-a)) \leq 0$, but $a(1-a) > 0$ for all $a \in (0, 1)$ — a contradiction.

P4b — wNSW Minimum Formula (UNSAT). The first-order condition $a^* C_R = (1 - a^*)C_W$ combined with the negation of $a^* = C_W/(C_R + C_W)$ is a polynomial system.

Z3 returns UNSAT in 6ms, confirming the formula algebraically for all $C_R, C_W > 0$.

State i *envies* state j if it prefers j's bundle scaled to its own population:

$$u_i\left(\frac{x_j^R \cdot n_i}{n_j}, \frac{x_j^W \cdot n_i}{n_j}\right) > u_i(x_i^R, x_i^W)$$

Note that scaling j's total bundle by n_i/n_j and dividing by n_i recovers j's *per-capita* quantities: $(x_j^R/n_j,\, x_j^W/n_j)$. Envy therefore asks whether state i would prefer j's per-capita bundle evaluated under i's own preferences.

PRO is envy-free. Under PRO, all states receive identical per-capita allocations: $x_i^R/n_i = M_R/\sum n_k$ and $x_i^W/n_i = M_W/\sum n_k$ for all i. Hence j's per-capita bundle equals i's own per-capita bundle and since all states evaluate bundles using the same Cobb–Douglas form, so the utility comparison yields equality. PRO satisfies envy-freeness under the scaled definition. Z3 verification: encoding this as a polynomial inequality with $n_i, n_j, M_R, M_W > 0$ should return UNSAT.

wNSW, Atkinson, and Leximin are not envy-free. Under these rules, per-capita bundles differ across states because allocations are tailored to preferences. Numerical verification finds 2 envy pairs under wNSW (Odisha envies Chhattisgarh; Telangana envies Andhra Pradesh), 245 pairs under Atkinson, and 416 pairs under Leximin. The Leximin case is instructive: although all $\tilde{u}_i$ are equal, a rice-preferring state such as Andaman & Nicobar Islands ($a_i = 0.812$) envies Bihar ($a_i = 0.526$) because Bihar receives a larger absolute rice allocation per capita, and evaluated under A&N's strong rice preference this yields higher utility. Z3 can confirm these with explicit witnesses.

7. Sensitivity Analysis

NE proxy uncertainty. Five Northeast states (Arunachal Pradesh, Manipur, Meghalaya, Mizoram, Nagaland) use Assam's value $a = 0.9537$ as a proxy; Tripura and Sikkim use $a = 0.8123$; Lakshadweep uses $a = 0.8758$. The Assam proxy likely overstates rice preference for Arunachal Pradesh, where wheat flour and noodle consumption is documented. A ± 0.05 shift in the Assam proxy moves the Leximin floor by ± 0.001 in $\tilde{u}$ units—aggregate metrics are robust, but individual NE state allocations (particularly wheat quantities under Leximin) are sensitive. A lower a_i for these states would increase their wheat allocation under preference-sensitive rules.

Reference year mismatch. H_i is from 2021 MPI; population is from 2024 Aadhaar. States with rapid poverty reduction between 2021 and 2024 may have slightly overstated n_i.

Case B (excluding DBT states). Removing Chandigarh and Puducherry from the allocation pool changes results by less than 0.001 in all metrics, as these states comprise

$< 0.03\%$ of total poor population as per data.

8. Limitations and Future Work

The model does not take into account urban consumption patterns, demand elasticity, and logistical constraints. Incorporating urban a_i (via HCES urban tables) would particularly affect Delhi, Maharashtra, and Karnataka, where urban-rural dietary patterns differ substantially. Second, the Leximin solution requires a dramatic reallocation of grain across FCI zones (e.g., diverting wheat from the Punjab-Haryana procurement base to the Northeast) which is a welfare benchmark, not an operational policy. Zonal logistics costs could be incorporated as a budget constraint in a more advanced model.

9. Conclusion

We modelled India's PDS grain allocation as a two-good fair division problem and compared four allocation rules across 36 states. The main findings are: (1) on the 16-state comparable subset, four states (Bihar, UP, Jharkhand, D&NH) robustly improve under every rule; a fifth (Madhya Pradesh) improves in the data as presented but is a borderline case under the 2011/2024 denominator correction; (2) PRO is simple but preference-blind; (3) wNSW maximises aggregate welfare but produces a systematic disadvantage for states with $a_i \approx 0.313$, derivable analytically; (4) Leximin achieves perfect equality at the cost of extreme preference-tailoring and logistical infeasibility.

The results suggest that a modified wNSW or Atkinson rule, which improve the worst-off states substantially over NFSA while remaining operationally closer to current practice may offer the best balance of fairness and implementability. Formal axiomatic verification confirms that PRO is envy-free under the standard scaled definition, while wNSW, Atkinson, and Leximin are not which is a consequence of preference-tailoring rather than a generic feature of the problem.

A. Statewise Welfare Comparison Against NFSA Baseline

Table 5 reports, for each of the 16 states with a two-good NFSA baseline, the per-state welfare under each rule and the worst-case gap $\underline{u}_i - \tilde{u}_i^{\text{NFSA}}$, where $\underline{u}_i = \min_r \tilde{u}_i^r$. A positive gap confirms improvement under every rule. The remaining 20 states receive a single grain type under NFSA (wheat = 0 or rice = 0), making the log-CD baseline utility undefined; they are omitted from this comparison but included in all allocation computations.

† Madhya Pradesh: after a +0.14 correction to the NFSA baseline for the 2011/2024 population denominator mismatch, the gap falls to approximately −0.04 under PRO — a borderline case.

Table 5: Per-state welfare $\tilde{u}_i$ under all rules vs. NFSA baseline, sorted by NFSA utility. $\underline{u}_i = \min_r \tilde{u}_i^r$. $\Delta_{\min} = \underline{u}_i - \tilde{u}_i^{\text{NFSA}}$. ✓ = improves under every rule.

State	a_i	**NFSA**	**PRO**	**wNSW**	**Atk.**	**Lex.**	$\underline{u}_i$	$\Delta_{\min}$	
D&NH & DD	0.329	−9.9164	−9.2134	−9.1522	−9.0642	−8.9788	−9.2134	+0.703	✓
Bihar	0.526	−9.8495	−9.0572	−9.0561	−9.0043	−8.9788	−9.0572	+0.792	✓
Jharkhand	0.681	−9.4557	−8.9336	−8.8682	−8.9011	−8.9788	−8.9788	+0.477	✓
Uttar Pradesh	0.378	−9.3768	−9.1744	−9.1433	−9.0568	−8.9788	−9.1744	+0.202	✓
Madhya Pradesh†	0.276	−9.3558	−9.2560	−9.1495	−9.0660	−8.9788	−9.2560	+0.100	✓
Gujarat	0.329	−8.9379	−9.2134	−9.1522	−9.0642	−8.9788	−9.2134	−0.276	×
Uttarakhand	0.378	−8.7310	−9.1744	−9.1433	−9.0568	−8.9788	−9.1744	−0.443	×
West Bengal	0.812	−8.5871	−8.8293	−8.6220	−8.7701	−8.9788	−8.9788	−0.392	×
Maharashtra	0.410	−8.4638	−9.1495	−9.1322	−9.0493	−8.9788	−9.1495	−0.686	×
A&N Islands	0.812	−8.1639	−8.8293	−8.6220	−8.7701	−8.9788	−8.9788	−0.815	×
Delhi	0.378	−8.0802	−9.1744	−9.1433	−9.0568	−8.9788	−9.1744	−1.094	×
Himachal Pradesh	0.094	−8.0505	−9.4003	−9.0152	−9.0097	−8.9788	−9.4003	−1.350	×
Jammu & Kashmir	0.718	−7.8416	−8.9046	−8.8087	−8.8691	−8.9788	−8.9788	−1.137	×
Ladakh	0.094	−7.6400	−9.4003	−9.0152	−9.0097	−8.9788	−9.4003	−1.760	×
Tamil Nadu	0.927	−6.7645	−8.7383	−8.3110	−8.6077	−8.9788	−8.9788	−2.214	×
Kerala	0.876	−5.6487	−8.7788	−8.4644	−8.6875	−8.9788	−8.9788	−3.330	×

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